

- 1 (a) The common woodlouse, *Oniscus asellus*, is a small invertebrate animal that is found in many areas of western and northern Europe.

Figure 1.1 shows some *O. asellus*.

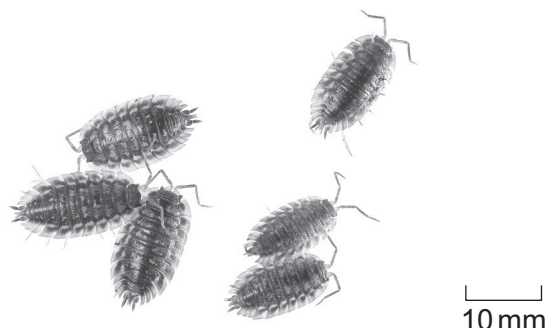


Figure 1.1

A student wanted to investigate the effect of temperature on the rate of respiration in *O. asellus*.

The student set up a respirometer as shown in Figure 1.2. A stand and clamp were used to hold the respirometer in position.

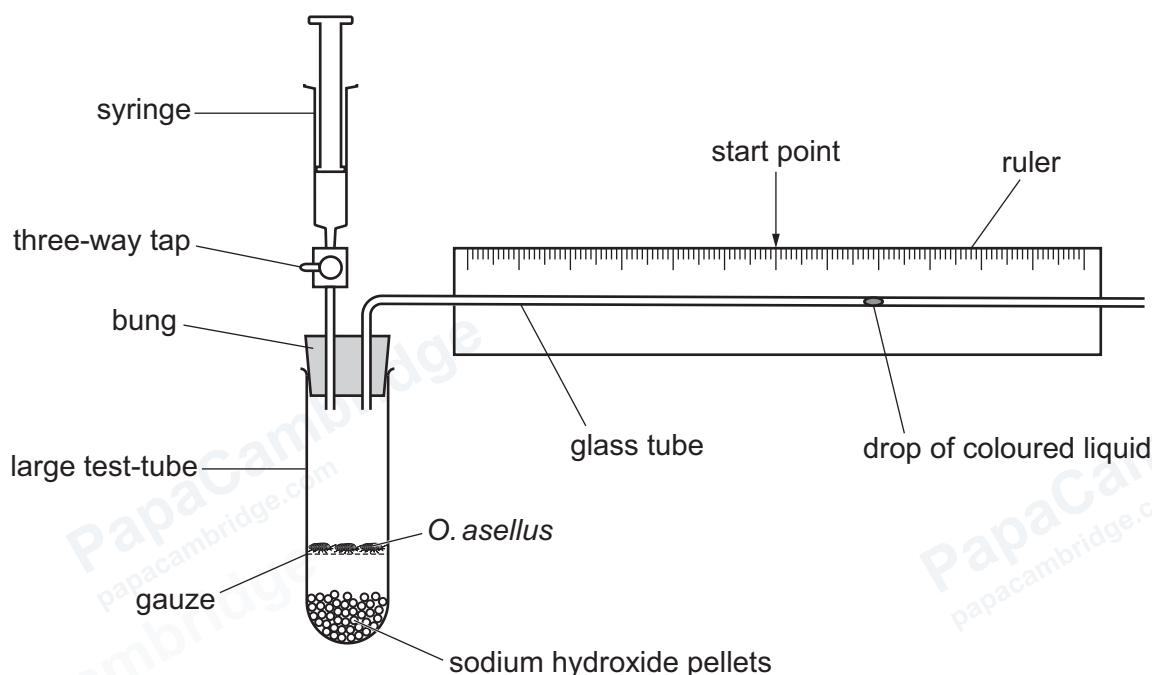


Figure 1.2

To determine the rate of respiration in *O. asellus*, the student:

- opened the three-way tap and used the syringe to move the drop of coloured liquid to the start point
- closed the three-way tap
- measured the distance moved by the drop of coloured liquid in a known time period.



- (i) Predict the direction of movement of the drop of coloured liquid during the investigation. Explain your answer.

direction

explanation

[1]

- (ii) Suggest a suitable control for this investigation.

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[1]

- (iii) Describe a method, using the respirometer shown in Figure 1.2, that the student could use to investigate the effect of temperature on the rate of respiration in *O. asellus*.

You should use temperatures from 5 °C to 35 °C, so that the *O. asellus* are not harmed in the investigation.

Your method should be set out in a logical order and be detailed enough to allow another person to follow it.

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[6]

- (iv) The student used the data, collected in the investigation, to calculate the rate of respiration in *O. asellus* at each temperature as a volume per unit time ($\text{mm}^3 \text{min}^{-1}$).

Describe how the student calculated the rate of respiration in units of $\text{mm}^3 \text{min}^{-1}$.

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- (b) The student then investigated the effect of different respiratory substrates on the rate of fermentation in yeast cells. The solutions of respiratory substrates were boiled to remove oxygen. The solutions were then cooled before mixing with the suspension of yeast cells.

A small test-tube was filled with 3 cm^3 of a yeast suspension and 2 cm^3 of a 20% glucose solution. The small test-tube was placed inside a large test-tube, as shown in Figure 1.3.

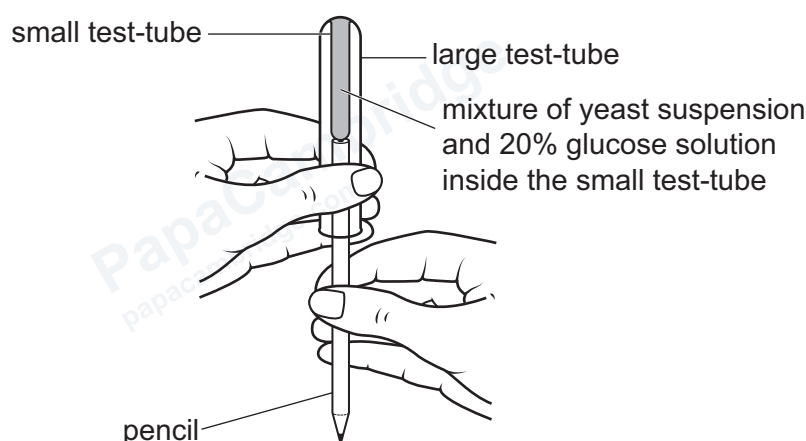


Figure 1.3

The test-tube was turned through 180° and placed in a test-tube rack for 30 minutes. During this time, gas was produced and collected as shown in Figure 1.4.

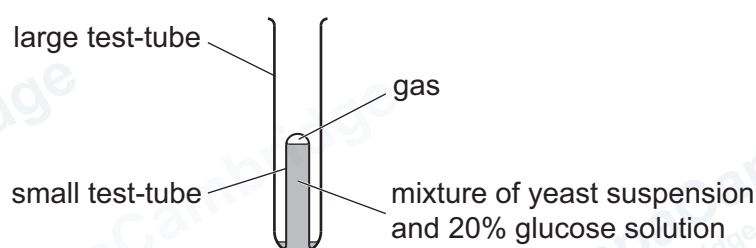


Figure 1.4

The student determined the volume of gas collected in the small test-tube after 30 minutes. This was used as a measure of the rate of fermentation.

The student repeated the experiment using a 20% maltose solution.

The student obtained data from five replicates using a 20% glucose solution and from five replicates using a 20% maltose solution.



- (i) Identify the **independent** variable in this investigation.

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- (ii) The student was provided with a 50% stock solution of glucose.

Describe how the student could use the stock solution to make **10 cm³** of a 20% glucose solution.

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- (iii) The student standardised the volume of the yeast suspension and the volume of the respiratory substrate solutions.

State **two other** variables that should be standardised in this investigation.

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(c) The results of the investigation are shown in Figure 1.5.

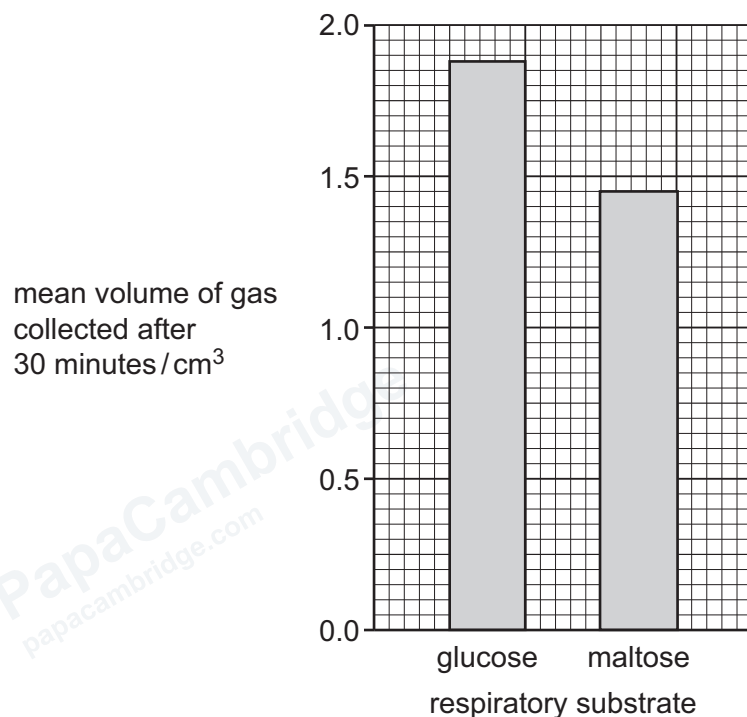


Figure 1.5

(i) Identify the type of data shown by the respiratory substrate in Figure 1.5.

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(ii) State a conclusion for the investigation, using evidence from Figure 1.5 to support your conclusion.

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- (iii) Suggest why the rate of fermentation in yeast cells is different with glucose as a respiratory substrate compared with maltose as a respiratory substrate.

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[Total: 17]